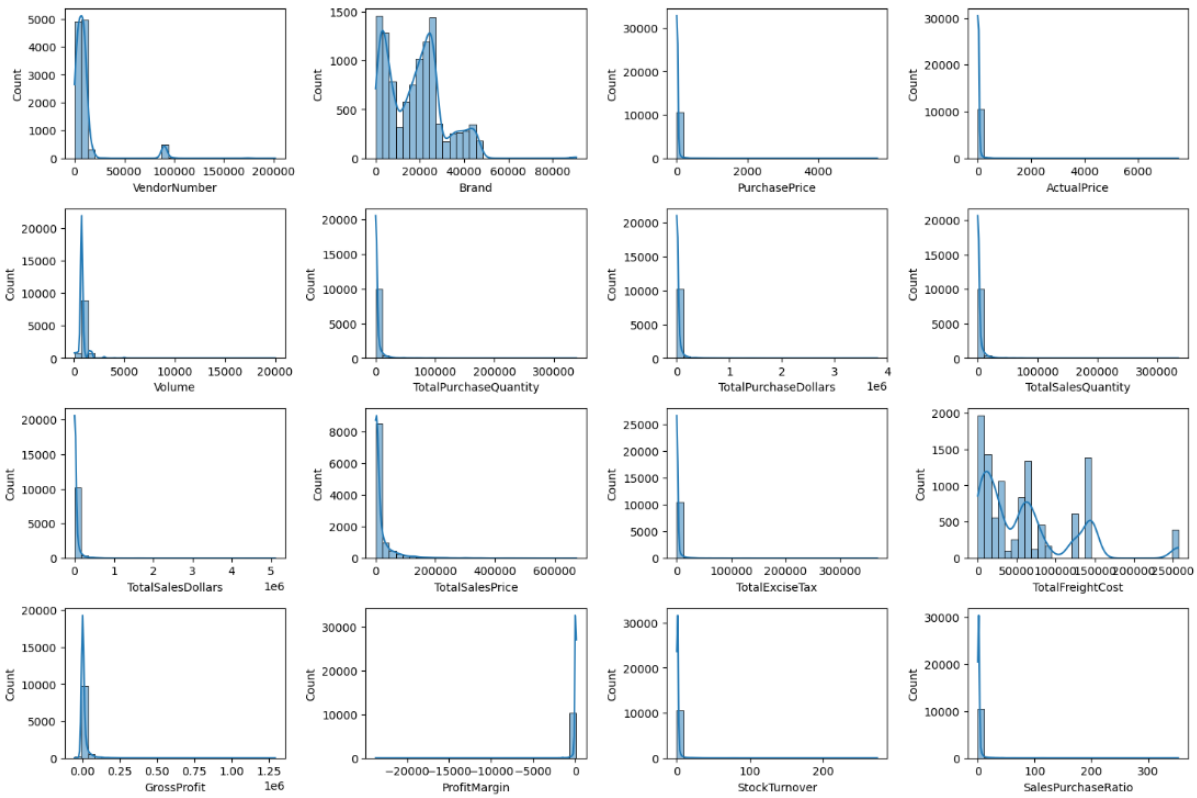
Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
TotalFreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02



Negative and zero values:

- **Gross Profit**: Minimum value is -52,002.78, indicating losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.
- **Profit Margin**: Has a minimum of (-infinity), Which suggests cases where revenue is zero or even lower than costs.
- **Total Sales Quantity and Sales Dollars**: Minimum value are 0, meaning some products were purchased bur=t never sold. These could be slow-moving or obsolete stock.

Outliers Indicated by high Standard Deviations:

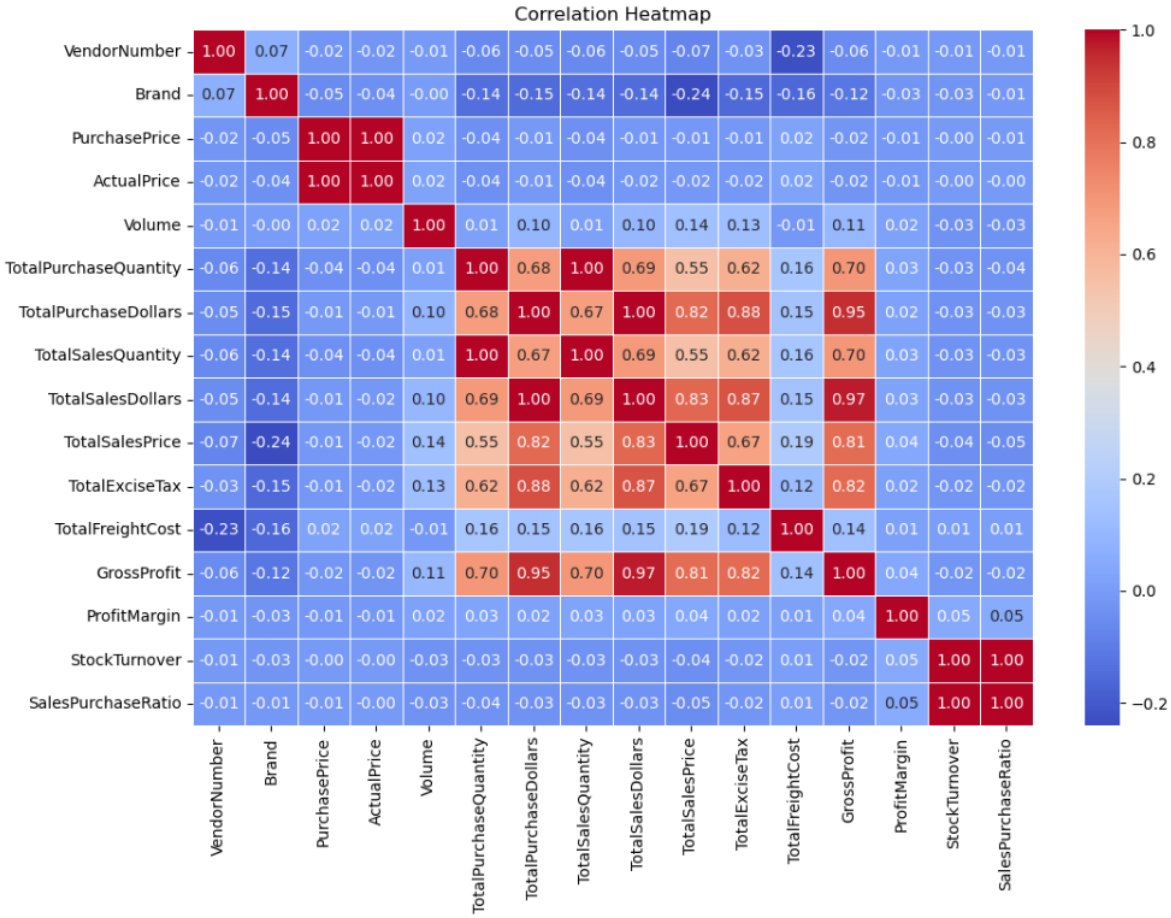
- **Purchase and Actual Prices**: The max values (5,680.81 & 7,499.99) are significantly higher than the mean (24.39 and 35.64), indicating potential premium products.
- **Freight Cost**: Huge variation, from 0.09 to 257,032.07, suggests logistics inefficiencies or bulk shipments.
- **Stock Turnover**: Ranges from 0 to 274.5, implying some products sell extremely fast while others remain in stock indefinitely. Value more than 1 indicates that sold quantity for that product is higher than purchased quantity due to either sales are being fulfilled from older stock.

Data Filtering

To enhance the reliability of insights, we remove inconsistence data where:

- Gross Profit < equal to 0 (to excludes transactions leading to losses).
- Profit Margin< equal to 0 (to ensuring analysis focuses on profitable transaction).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).

Correlation Insights



- **Purchase Price vs. Total Sales Dollars & Gross Profit**: PurchasePrice has weak correlation with TotalSalesDollars(-0.012) and GrossProfit(-0.016), suggesting that price variations do not significantly impact sales revenue or profit
- **Total Purchase Quantity vs. Total Sales Quantity**: Strong correlation between total purchase quantity and total sales quantity (0.999), confirming efficient inventory turnover.
- **Profit Margin vs. Total Sales Price**: Negative correlation between profit margin and total sales price(-0.179 suggests that as sales price increases, margins decrease, possibly due to competitive pricing pressures.
- **Stock Turnover vs. Gross Profit & Profit Margin**: StockTurnover has weak negative correlation with both GrossProfit(-0.038) and ProfitMargin(-0.055), indicating that fater turnover does not necessarily result in higher profitability

Research Questions & Key Findings

1. Brands for Promotional and Pricing Adjustments

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Brands with Low Sales but High Profit Margins:
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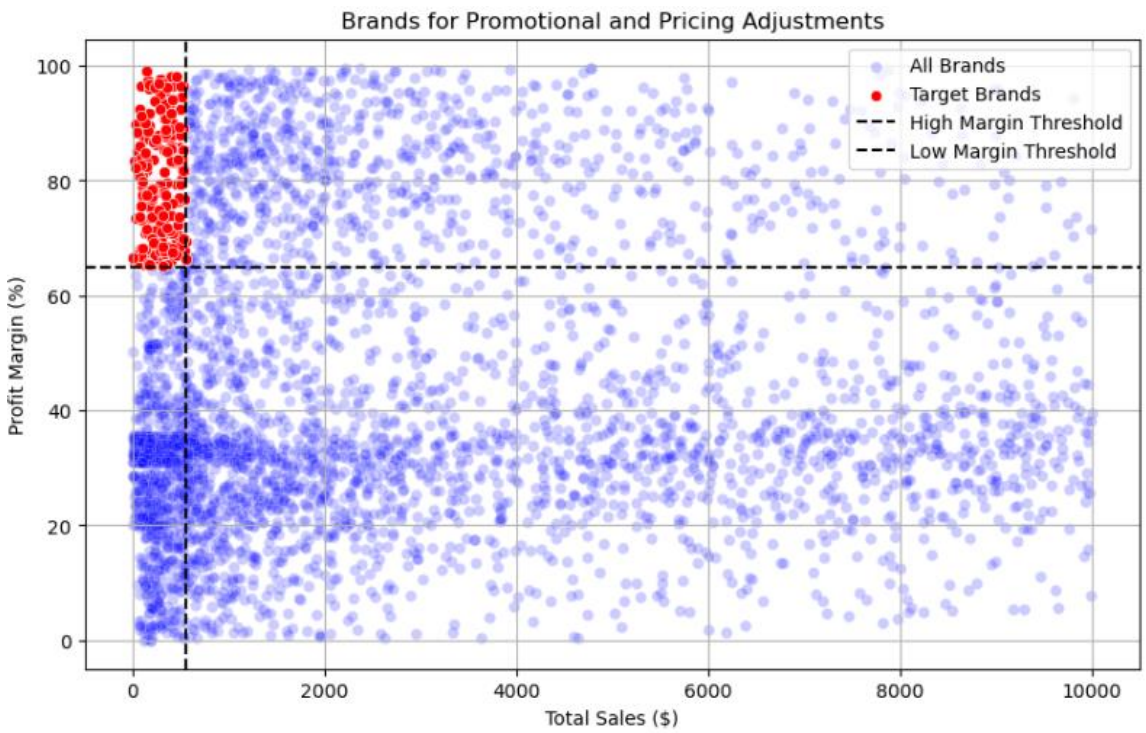
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

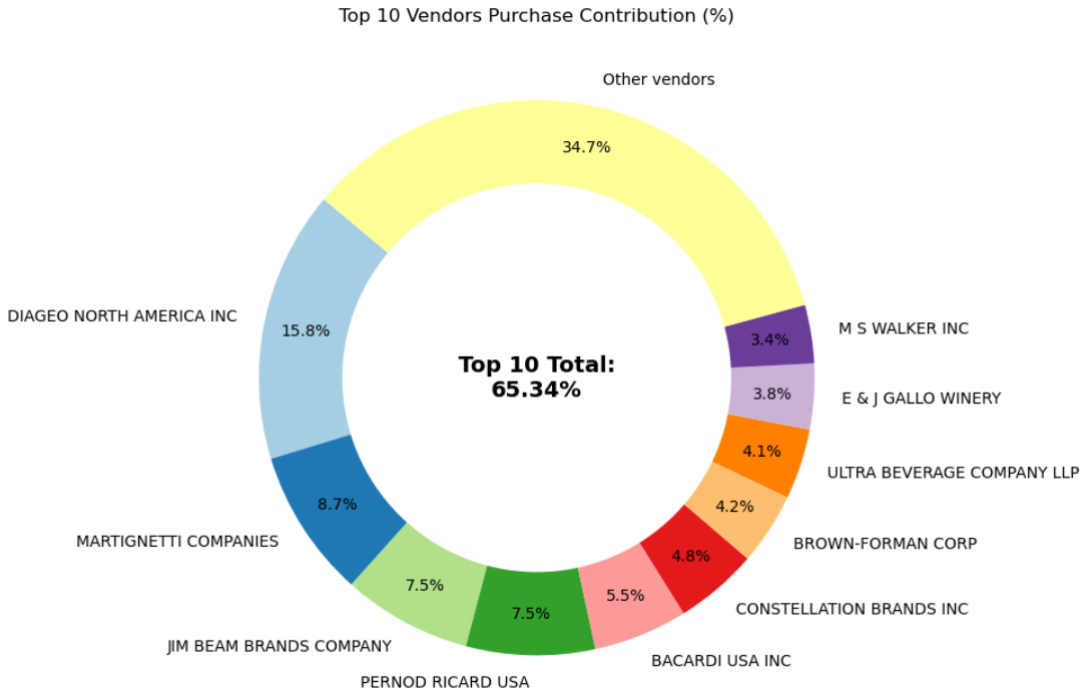
198 rows × 3 columns

198 brands exhibits lower sales but higher profit margins, which could benefits from targeted marketing, promotions or price optimizations to increase volume without compromising profitability.



2.Top Vendors by Sales and Purchase Contribution

The top 10 vendors contribute 64.34% of total purchases, While the remaining vendors contribute only 34.7%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions.



3.Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receives a 72% lower unit cost ($10.78 unit vs. higher unit cost in smaller orders).

Bulk pricing strategies encourage larger orders increasing total sales while maintaining profitability.

OrderSize	UnitPurchasePrice
Small	39.068186
Medium	15.486414
Large	10.777625

4.Identifying Vendors with Low Inventory Turnover

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Total Unsold Capital: 2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and
Affects overall profitability.
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	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21k
46	JIM BEAM BRANDS COMPANY	554.67k
68	PERNOD RICARD USA	470.63k
116	WILLIAM GRANT & SONS INC	401.96k
30	E & J GALLO WINERY	228.28k
79	SAZERAC CO INC	198.44k
11	BROWN-FORMAN CORP	177.73k
20	CONSTELLATION BRANDS INC	133.62k
61	MOET HENNESSY USA INC	126.48k
77	REMY COINTREAU USA INC	118.60k

5. Profit Margin Comparison: High vs. Low-Performing Vendors

Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17%

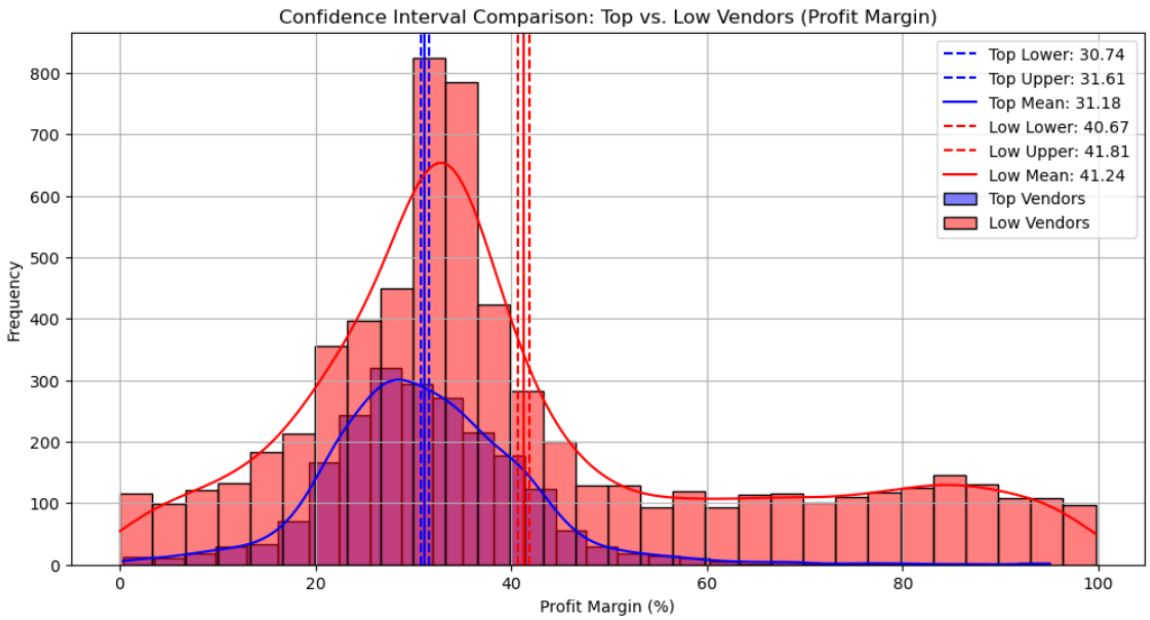
Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

a) Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.

b) Low performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



Final Recommendations

- Re-evaluate pricing for low-sales, high margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.